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The Educational Buffer:

Investigating the Interaction Between Formal Education and Structural Brain Atrophy in Alzheimer's Disease

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Abstract

This study investigated the relationship between years of educational attainment and brain atrophy in the context of dementia patients using secondary quantitative data analysis. Using secondary data from a cross-sectional sample of 436 participants from ages of 18 to 96, statistical analyses including linear regression and analysis of variance (ANOVA), with post-hoc comparisons were conducted. Results indicated that among individuals with dementia, education was associated with greater mean brain volume, which suggested that educational attainment acts as a protective factor against dementia-related atrophy. However, these results also indicate that education does not prevent brain atrophy. Overall, these findings support the role of educational attainment in providing protection against structural decline, rather than eliminating brain atrophy caused by dementia.

Keywords**:** dementia, cognitive reserve, normalized whole brain volume (nWBV), neurodegeneration, Educational Attainment, Clinical Dementia Rating

Introduction

Dementia is a growing concern, with the World Health Organization (2021) estimating that approximately 57 million people worldwide are living with dementia. Dementia is a neurological condition that refers to a significant cognitive decline that interferes with daily functioning (Christova et al., 2025). It should be noted that it is not classified as a specific disease, but instead as a syndrome, which means that it describes a cluster of symptoms that often occur together (Gale et al., 2018). The collective symptoms that dementia describes include impairments across cognitive domains, such as memory, language, attention and executive

function (Cipriani et al., 2020). These symptoms often result from underlying degenerative neurological conditions such as Alzheimer's, vascular dementia, Lewy body disease, and frontotemporal dementia (Amin & McCausland, 2024). Several risk factors have been associated with dementia in multiple domains of life. These domains include lifestyle behaviours (physical inactivity, obesity,excessive alcohol consumption and smoking), cardiovascular domains (hypertension and diabetes), cognitive and social domains (depression, less education and social isolation), and environmental and injury domains (traumatic brain injury and pollution) (Livingston et al., 2024). While dementia is caused by a variety of different factors, education has been thought to provide a protective factor for developing dementia (Meng & D'Arcy, 2012). This is because education is thought to be one type of life experience that contributes to cognitive reserve (CR), a factor that helps maintain cognitive functioning despite age-related decline (Liu et al., 2024). CR is said to influence the efficiency and capacity of brain networks, which is why it is thought to protect from neurological damage (Stern, 2009). While CR may explain how individuals maintain cognitive function, dementia is also characterized by structural changes in the brain, such as progressive brain atrophy (Lv et al., 2025). Brain atrophy refers to the reduction in brain volume (Sungura et al., 2021). Research across different types of dementia has shown that brain atrophy affects multiple brain regions rather than a single localized area. These areas often include the hippocampus, as well as the frontal and temporal lobes (van de Pol et al., 2006). Studies regarding educational impact on cognitive decline have produced mixed findings. Some research suggests that education provides a protective effect from cognitive impairment, however, this protection factor decreases with age (Zhong et al., 2024). Additionally, research specifically focused on brain atrophy suggests that high levels of educational attainment is associated with slower cognitive decline, however the level of

education makes individuals more sensitive to atrophy once it has started (Mungas et al., 2018). These findings suggest that although education may protect against the onset of cognitive symptoms, it may not prevent brain degeneration. Research examining how CR affects the brain supports this claim by showing that CR is associated with better cognitive functioning despite greater levels of brain atrophy (Pettigrew & Soldan, 2019). Although education has been studied as a protective factor through CR, there is still limited research regarding whether educational attainment predicts the degree of brain atrophy in individuals with dementia. Prior research primarily focused on cognitive function, rather than if educational attainment could protect the brain from structural changes from dementia. This raises the question of whether years of educational achievement predict the degree of physical brain atrophy in clinical dementia patients. Using secondary quantitative analysis, this study aimed to examine whether years of educational achievement can predict the degree of physical atrophy of whole brain volume (nWBV) in clinical dementia. Based on previous research, it was hypothesized that higher education would serve as a structural buffer. Understanding whether educational attainment contributes to resilience against structural changes associated with dementia may provide important insight into the progression of the disease.

Methods

This study employed a quantitative, observational research design, utilizing secondary data analysis from the Open Access Series of Imaging Studies (OASIS) cross sectional MRI-dataset (N=350), a well-established and publicly available dataset focused on brain imaging, aging, and dementia.

The primary objective of the study was to investigate the measurement of normalized whole brain volume (nWBV) in individuals across a wide range of ages and varying stages of clinical dementia. By examining these relationships, the study aimed to better understand how brain atrophy correlates with both aging and cognitive decline through stages of dementia, as well as how factors such as education may influence these patterns. Gender and socioeconomic (SES) status were addressed as potential cofounders.

The sample consisted of participants ranging in age from approximately 18 to 96 years, allowing for a comprehensive analysis across the wide adult lifespan. This broad age distribution provided an opportunity to observe both early and late changes in nWBV. Participants were also categorized based on their Clinical Dementia Rating (CDR), which ranged from 0 to 1 within the datasets observed. Participants categorized as CDR 2 (moderate dementia) were excluded because of a low sample size available. A CDR score of 0 indicates no signs of dementia, while a score of 1 reflects mild dementia. By assessing individuals across this spectrum, the study was able to compare nWBV and related variables between non-demented and demented participants.

In addition to age and dementia status, years of education were included as an important variable in the analysis. Education is often considered a proxy for cognitive condition (Liu et al., 2024), a concept suggesting that individuals with higher levels of education may be better able to cope with or compensate for brain changes associated with aging or dementia. As such, examining education alongside brain volume and dementia severity provided further insight into an assessment of whether education plays a role in the development of dementia. Participants in the examined studies reported education ranging from approximately 6 to 23 years.

Brain atrophy, a key indicator of neurodegeneration, was quantified using nWBV. This measurement allowed for standardized comparisons across individuals by accounting for

differences in head size, total intracranial volume, and differences in grey and white matter present in the brain. By using nWBV, the study ensured that variations in brain volume were reflective of actual atrophy rather than natural differences in anatomy across participants. Measurements of nWBV were then analyzed in relation to both age and dementia severity to identify any significant trends or patterns present among participants.

Clinical dementia severity was assessed using the Clinical Dementia Rating (CDR) scale, a widely used clinical tool that evaluates cognitive and functional performance in areas such as memory, orientation, judgment, problem solving, personal care, hobbies, and daily activities. (Knight Alzheimer Disease Research Center, 2026)

The use of CDR allowed for a consistent and reliable classification of participants into different dementia stages, making it possible to compare brain volume across groups and visualize trends. Statistical analyses were conducted using RStudio to evaluate the relationships among education, dementia status, and brain volume. Differences in years of education across dementia stages were examined by comparing individual participant data, allowing for the identification of any patterns linking education level to cognitive status.

To further support the analysis and enhance interpretability, several types of data visualizations were generated. An Analysis of Variance (ANOVA) compared the means of three different independent groups including demented vs non demented, years of education, and nWBV to determine whether a specific mean presented itself as unusually high or low in order to assess the relationship in determining a rate of dementia present in participants. Group differences in brain volume between demented and non-demented individuals were analyzed to assess the impact of dementia on brain atrophy.

Scatter plots were used to illustrate the relationship between age and brain volume, helping to identify trends such as age-related decline in nWBV. As well as the relationship between nWBV and years of education. Box plots were used to compare distributions of education across dementia stages and to visualize differences in brain volume between demented vs non demented groups.

These graphical representations allowed for clearer identification of patterns, variability, and potential outliers within the data.

Overall, this study combined quantitative methods and visual analysis techniques to explore the complex relationships between aging, brain volume , education, and dementia. By leveraging an existing dataset, it provided valuable insights into how these variables interact and contribute to the diagnosis of dementia across lifespan, and how factors such as education play a role in the diagnosis.

Results

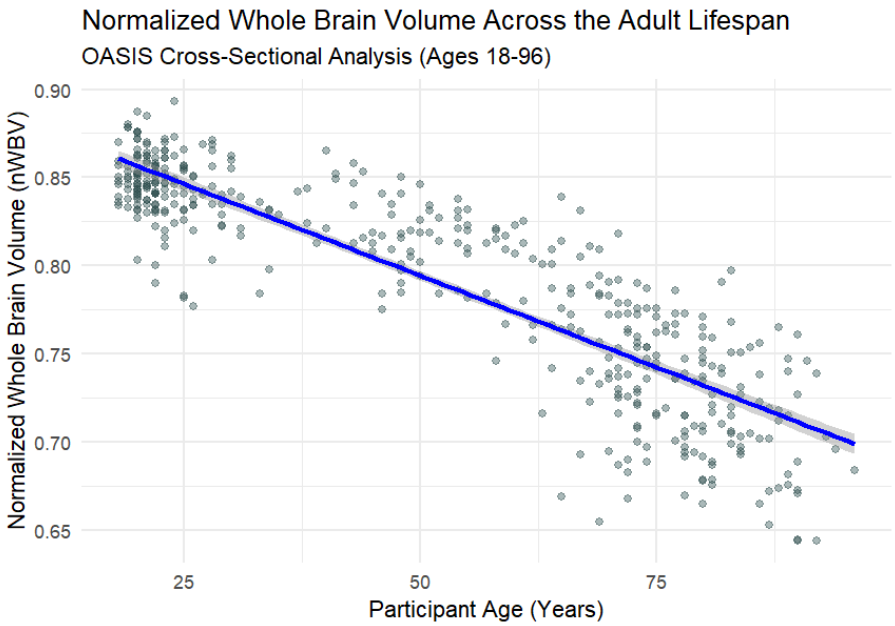


Figure 1. Baseline normalized whole brain volume (nWBV) that predictably declines as a function of age across the adult lifespan ($p < 0.001$, $R^2 = 0.764$).

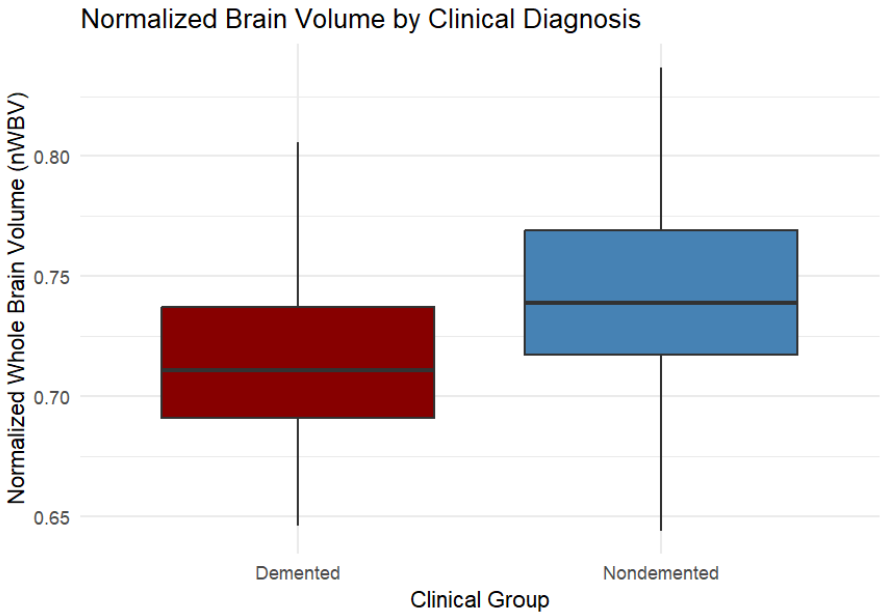


Figure 2. Boxplot comparing physical brain volume of participants categorized by clinical diagnosis. One way ANOVA confirmed a highly significant reduction in nWBV with demented compared to nondemented participants ($p < 0.001$, $\eta^2 = 0.101$). These results validate that clinical cognitive decline in the population is accompanied by physical brain atrophy.

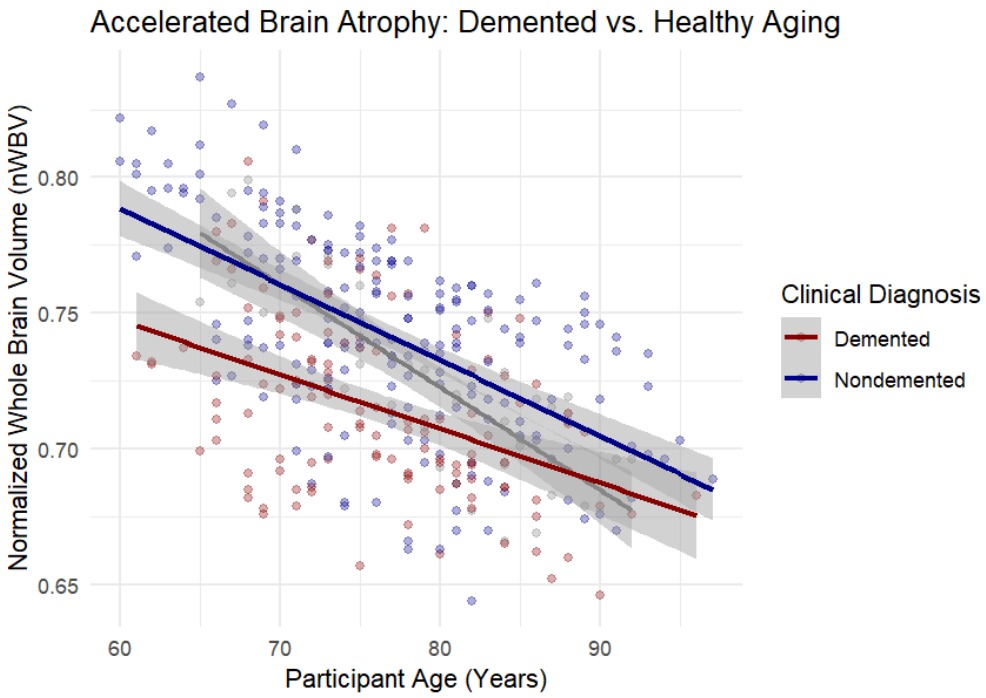


Figure 3. Plot that trajects brain volume loss (nWBV) between demented and nondemented seniors. While age related decline is universal, the demented cohort (red line) shows a significantly lower volume compared to healthy cohorts (blue line), $F(2, 348) = 19.6$ and $p < 0.001$. The widening gap represents the acceleration of brain volume atrophy caused by neurodegeneration.

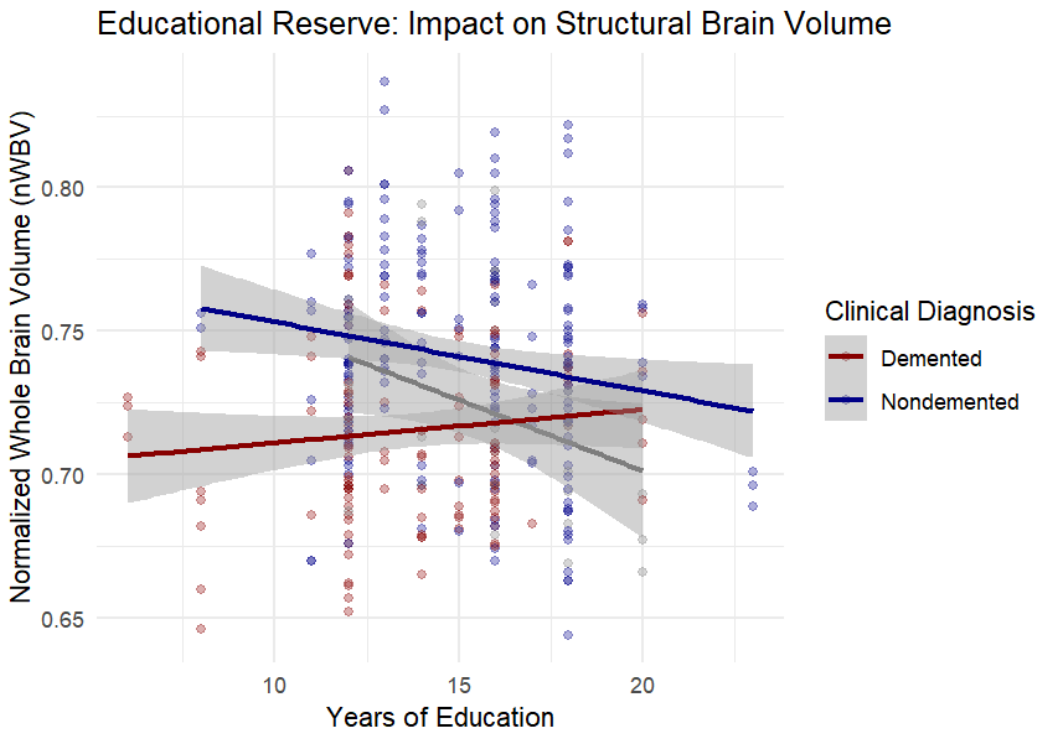


Figure 4. This plot demonstrates how years of formal education correlate with brain volume. The distinct regression slopes for both groups, illustrating that higher educational attainment may provide a buffer to brain atrophy as a clinical symptom ($p = 0.0115$, $\eta^2 = 0.067$).

A two way ANOVA was conducted to examine the effects of educational attainment and clinical diagnosis on normalized whole brain volume. Assumptions of normality and homogeneity of variance were assessed and met. Analysis began by establishing a baseline for brain atrophy across varying ages. Data revealed a negative correlation in decline of nWBV as a function of age from 18 to 96 years old (Figure 1). The model generated a p value less than 0.001, indicating that chronological age is a consistent predictor of brain atrophy in the general population. Age was analyzed separately due to its strong association with brain volume and was considered a key confounding variable when interpreting the relationship between education and

nWBV. Additionally other potential confounding variables including socioeconomic status, sex, and quality of education were not controlled for in the analysis

A one way ANOVA measured the difference between brain volume between clinical cohorts. Results confirmed a reduction in nWBV with participants diagnosed with dementia compared to nondemented patients (Figure 2). This comparison produced a statistically significant difference in brain tissue between the two groups, as indicated by a p-value of 1.48×10^{-9} and an effect size of $\eta^2 = 0.11$.

An overlay analysis compared the rate of brain volume loss between clinical groups. While both cohorts (demented and nondemented) exhibited a negative correlation between age and nWBV, the demented group demonstrated a statistically significant lower volume baseline throughout the senior lifespan as indicated by an F-value and p-value of 19.6 and less than 0.001 respectively (Figure 3). The final primary analysis looked at the relationship between the amount of formal education and nWBV. A two way ANOVA yielded a statistically significant interaction between educational attainment and clinical diagnosis on brain volume ($p = 0.0115$, $\eta^2 = 0.067$). This was visualized through regression slopes of each cohort (Figure 4).

Additional analysis regarding the distribution of cognitive performance with different lifestyle factors are detailed in the appendix section. These include the distribution of cognitive scores across dementia stages of CDR 0 to 1 (Figure 5), the impact of socioeconomic status on cognitive screening scores (Figure 6), and the variance of educational years across specific clinical ratings of 0 to 1 (Figure 7). Finally, Figure 8 exhibits the general distribution of nWBV across all recorded formal educational years (6 to 23).

Discussion

While there is not any way to fully prevent dementia, certain approaches may help reduce risk factors of dementia that can be started in early life. There are specific risk factors that suggest that education plays an important role in dementia risk and long term education has been associated with the prevention against dementia (Livingston et al., 2024). Cognitive reserves are said to be the brain's defense mechanisms, which may act as a buffer against structural changes caused by neurological disorders (Liu et al., 2024). The large amount of cortical thickness that is made by support cells called glia, compensate for brain damage. Another kind of reserve is brain reserve, which is when individuals with larger brain volumes may be able to tolerate greater levels of brain changes and may result in decreasing the chances of getting Alzheimer's Disease in the early stages (Liu et al., 2012). Figure 5 compared the physical brain volume of demented and nondemented patients with physical brain atrophy. However education plays a significant role when it comes to reducing cognitive decline (Figure 6). From 6 to 23 years of education, demented patients nWBV are closely related to nondemented patients. This may reflect the brain compensating early through a potential mechanism of structural buffer. As years of formal education increase, more connections may be formed in the brain, contributing to the development of more neural pathways (Livingston et al., 2024). This may decrease the likelihood of cerebral atrophy and ultimately Alzheimer's Disease at early ages. However, Alzheimer's Disease is often studied by examining neuroimaging techniques, which has shown that affected individuals exhibited reduced cortical thickness. Changes in cortical thickness may also be caused by the body's response to inflammation. This problem can present a misdiagnosis of the early signs of Alzheimer's Disease. In addition to the potential of misdiagnosis there is less awareness of early on-set dementia, some individuals with early on-set dementia syndrome may

not seek medical treatment. This may influence the figures for dementia prevention, especially for data talking about early on-set dementia prevention. Talking about Alzheimer's Disease with the use of Transcranial Magnetic Stimulation (TMS) which can be used to identify the early signatures of synaptic dysfunction associated with Alzheimer's Disease (Koch & Spampinato, 2022). There is evidence to support the hypothesis that loss of synaptic density likely precedes neuronal degeneration, meaning that the impairment of neuroplasticity mechanisms could play a big role in progressive cognitive impairments (Koch & Spampinato, 2022). The altered mechanisms of neuroplasticity have been linked to different types of neural changes including neural network disarrangement, and cell death suggesting synaptic dysfunction may be a driver of Alzheimer's Disease -related cognitive decline (Koch & Spampinato, 2022). The downside is that there is no way to accurately measure synaptic function or dysfunction directly in Alzheimer's Disease patients, but they can measure total brain activity using TMS which allows for real time modifications of the electric activity of the brain caused by stimulation (Koch & Spampinato, 2022). TMS can induce electric current in the brain which triggers the depolarization of cortical neurons, and if done repeatedly stimulation induces progressive changes in brain activity which could allow for TMS to overcome the limitations of imaging techniques to track synaptic dysfunction in incipient dementia (Koch & Spampinato, 2022). For initial prevention of dementia there is strong evidence showing that being physically, socially, cognitively active in midlife (aged 18-65) and during late life (aged older than 65), could give greater dementia prevention and can maintain cognitive ability even with individuals with less education (Livingston et al., 2024). Addressing the twelve risk factors mentioned earlier may be associated with up to a 40% reduction in dementia risk (Livingston et al., 2024). Talking more specifically about Alzheimer's disease, the greatest risk factor is from family history with there

being a 3-fold higher risk factor if your first-degree relatives (parents, siblings, children) are patients with Alzheimer's Disease (Amin & McCausland, 2024). Direct support for the genetic component of Alzheimer's Disease comes from the mutation of three genes (amyloid precursor protein, presenilin 1, presenilin 2) that lead to early-onset (aged around 65) Alzheimer's Disease (Amin & McCausland, 2024). Although there is no cure for Alzheimer's Disease, research for early diagnoses have started to be studied. Abnormal protein formation such as clumping and tangles of amyloid beta proteins are scouted for diagnoses of Alzheimer's Disease; furthermore, recent experimentation of new biomarkers are also being developed. Blood and plasma based biomarkers have been explored as the alternative less invasive method of diagnosis than CSF biomarker tests (Wells & Horton, 2022). Blood tests may allow for tracking of abnormal biomarkers; however, these blood based markers only focus on amyloid and tau proteins and not on other neurodegeneration biomarkers, (Wells & Horton, 2022). But, these biomarkers have not been too consistent compared to CSF markers for early diagnosis, because they are not always specific for Alzheimer's Disease (Wells & Horton, 2022). The challenges with blood based biomarkers are that blood contains lower concentration of the markers than CSF. Another challenge is that blood is complex and has other components which can ultimately influence biomarker levels (Wells & Horton, 2022). Other biomarkers that are currently being more researched on are salivary, olfactory and retinal and ocular changes (Wells & Horton, 2022).

Conclusion:

Although the cure for dementia seems uncertain, the research for early diagnostic measures are becoming more evolved. This study demonstrated that education may act as a significant structural buffer against dementia. The observed interaction with education and

clinical diagnosis ($p = 0.0115$) suggests that higher education levels may help maintain neural architecture and delay symptoms. Furthermore, new emerging technologies like blood based biomarkers and Transcranial Magnetic Stimulation (TMS) are being developed that may help detect dementia early on. Ultimately, these findings highlight the value of early life educational investment that may help as a lifelong protection against brain atrophy. By shifting the focus toward building structural resilience instead of solely treating decline, such findings could create a foundation for improving long term outcomes in aging populations.

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Appendix

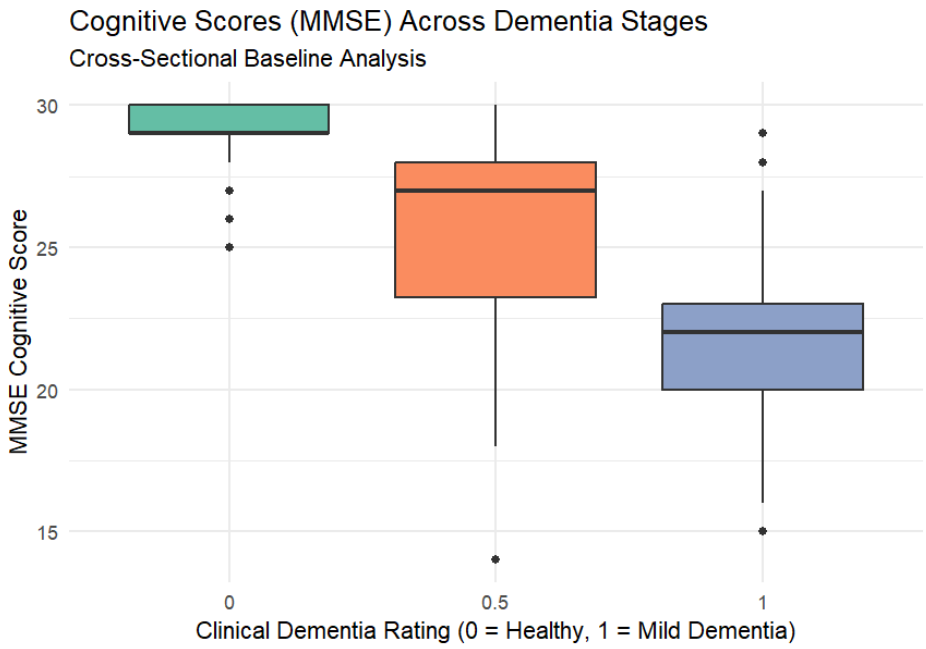


Figure 5. This boxplot illustrates the age distribution across different stages of impairment, with healthy individuals (CDR 0) and those with very mild decline (CDR 0.5) sharing similar median ages. The "moderate" stage (CDR 2) is mostly composed of significantly older participants. ANOVA test indicated a statistical significance in MMSE cognitive score across varying CDR scores.

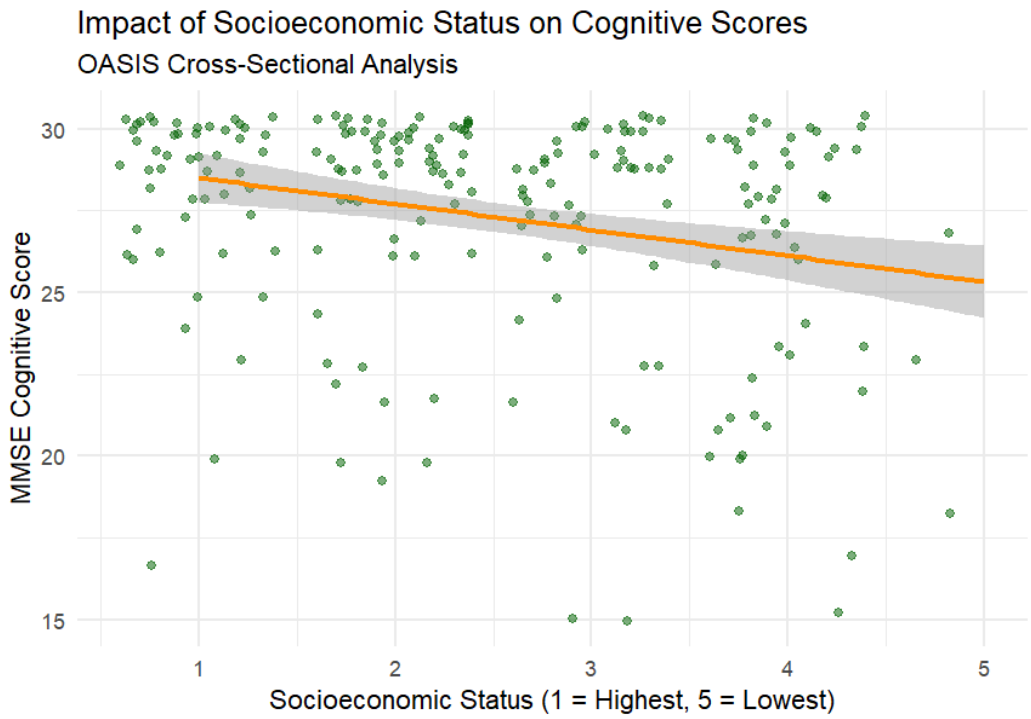


Figure 6. This scatter plot illustrates the impact of socioeconomic factors on cognitive health across 216 participants. Using a linear regression model, a highly significant negative correlation was found between Socioeconomic status (1 represents highest and 5 represents lowest status) and MMSE cognitive scores ($p < 0.001$).

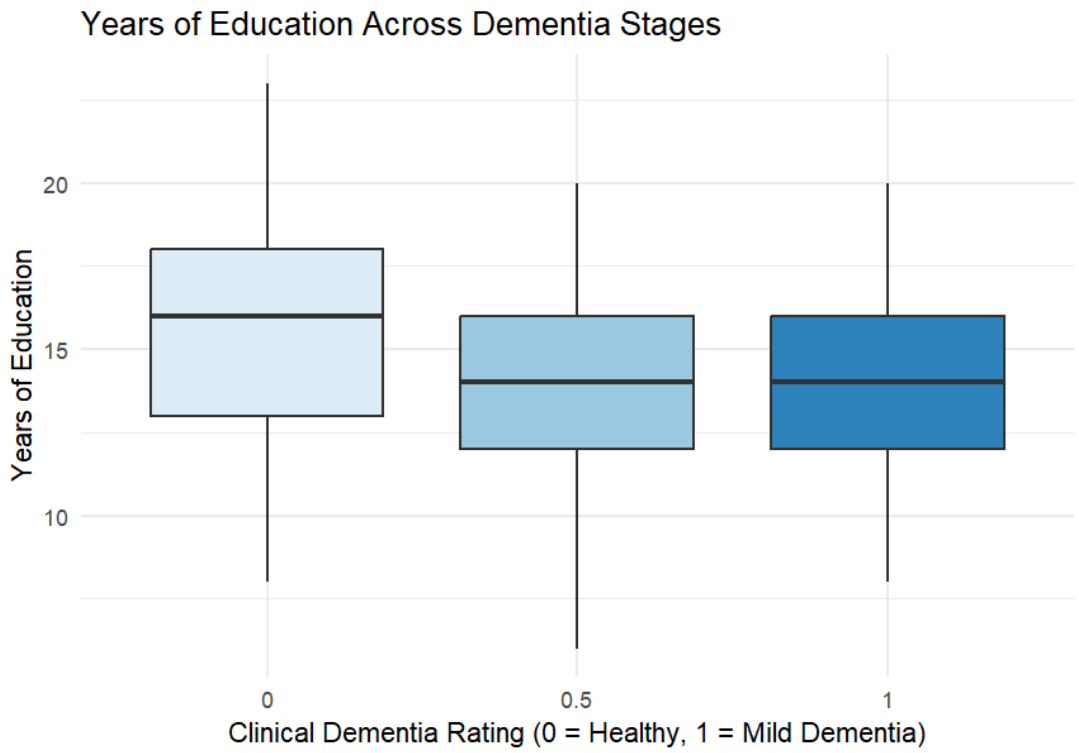


Figure 7. Boxplot of relationship between years of formal education (EDUC) categorized by clinical dementia rating (CDR), ranging from healthy controls (CDR 0) to mild dementia (CDR 1). ANOVA test confirmed significant differences in education across groups ($p = 0.001$). Post-hoc testing indicated that the most significant differences occurred between healthy individuals and those with mild dementia ($p = 0.002$). While mild dementia (CDR 1) also showed lower mean education compared to healthy controls, where the comparison did not reach statistical difference in this dataset ($p = 0.067$).

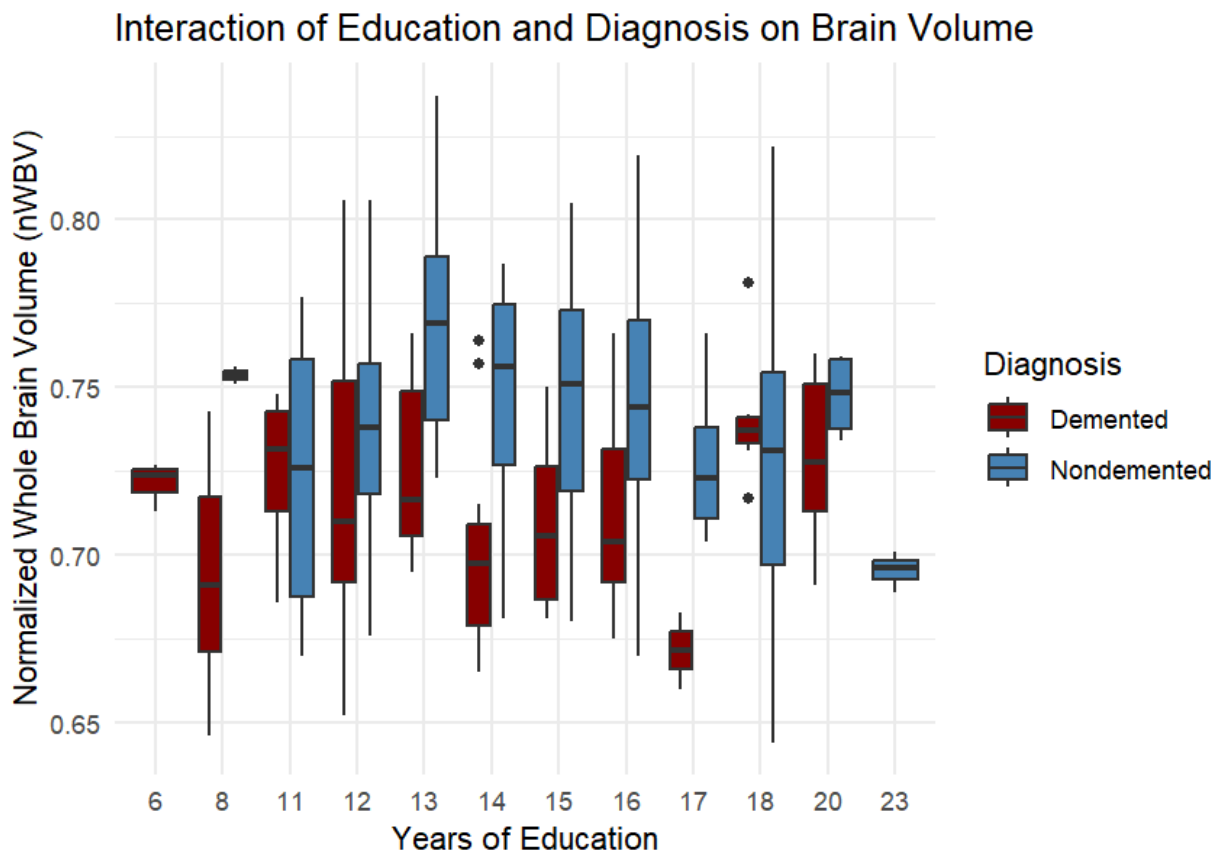


Figure 8. Distribution of nWBV across years of formal education (6 to 23 years). Two way ANOVA confirmed statistically significant association between education and nWBV ($p = 0.0089$). Results provide structural evidence that educational attainment is linked to brain volume, suggesting that higher education contributes to a higher reserve of brain tissue that buffers cognitive decline.

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